

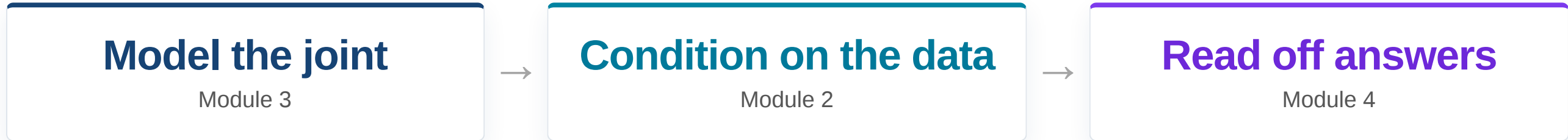
Probabilistic Machine Learning

Lecture 02: Probability and Statistics

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Where We Left Off

Lecture 01 ended on three moves, and promised that their vocabulary would be made precise here:



It also left a claim unpaid: that **epistemic** uncertainty is the kind more data removes. Module 4 turns that slogan into a rate.

Module 1

A Language for Uncertainty

Guiding Question: *We need a language that can say "how strongly", not just "true or false". What must such a language satisfy — and who decides the numbers?*

Example: Two Fruit Boxes

Red box

2 apples, 6 oranges

Blue box

3 apples, 9 oranges

We pick a box, then a fruit from it.

- If we pick the **red box** **40%** of the time, how often do we get an apple?
- We got an **orange** — which box did it probably come from?

The Counts, in One Grid

Twenty picks, recorded as a table of counts — box across, fruit down:

	$B = r$	$B = b$	row total
$F = a$ (apple)	2	3	5
$F = o$ (orange)	6	9	15
column total	8	12	20

Joint

a single cell over the total

$$P(B = r, F = a) = \frac{2}{20}$$

Marginal

a row or column total over the total

$$P(F = a) = \frac{5}{20}$$

Conditional

a cell over its own row or column

$$P(F = a \mid B = r) = \frac{2}{8}$$

The Two Rules

Sum rule

$$P(X) = \sum_Y P(X, Y)$$

Add along a row or column to remove a variable.

Product rule

$$P(X, Y) = P(Y | X)P(X)$$

A cell is its column total times its share of that column.

Almost all of probability theory is these two rules applied repeatedly.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y | X) = \frac{P(X | Y)P(Y)}{P(X)}, \quad P(X) = \sum_Y P(X | Y)P(Y)$$

Which box did the orange come from?

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \quad P(X) = \sum_Y P(X \mid Y)P(Y)$$

Which box did the orange come from?

$$P(B = r \mid F = o) = \frac{P(F = o \mid B = r)P(B = r)}{P(F = o)} = \frac{\frac{6}{8} \cdot 0.4}{0.75 \cdot 0.4 + 0.75 \cdot 0.6} = 0.4$$

Here the fruit told us **nothing** about the box — both boxes have the same orange fraction. Evidence is only informative when the hypotheses predict different things.

Takeaway

Two rules — **sum** to remove a variable, **product** to split a joint — generate joint, marginal, conditional probabilities and Bayes' theorem. The rest of this chapter makes them precise and extends them beyond counting.

What Makes an Experiment "Random"?

Definition (Random experiment). A procedure that can — at least in theory — be repeated infinitely often, characterised by:

1. **Configuration** — a precisely specified system that can be reconstructed.
2. **Procedure** — an instruction for executing the experiment.
3. **Unpredictability** of the outcome.

Coin flip: a 1-Euro coin (configuration), placed on the thumb and snapped into the air (procedure), record heads or tails (outcome).

Randomness as Missing Information

A random experiment is still causal. Its unpredictability reflects **missing information about the causal chain**, not an absence of causes.

Consequence: a random experiment can become a deterministic process once new information is available.

A coin flip filmed at 10,000 fps and measured precisely enough stops being random. What does that imply for a model that calls something "random"?

Sample Space and Events

Definition (Sample space, event space). A set $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$ is the **sample space** of a random experiment if every outcome is associated with exactly one element $\omega \in \Omega$. The elements are called **outcomes**.

Let Ω be finite. Every subset $A \subseteq \Omega$ is an **event** the event A **occurs** iff the observed outcome ω is a member of A . The set of all events, the power set $\mathcal{P}(\Omega)$, is the **event space**.

An outcome is a single result; an event is a *question you can ask* about the result.

For infinite Ω — the drug dosage below — the event space is a suitable family of subsets rather than the full power set we return to this when continuous distributions appear.

Important Event Types

$$\emptyset$$

impossible event

$$|A| = 1$$

elementary event

$$\Omega$$

certain event

$$A \subseteq B$$

A is a sub-event of B — " A entails B "

$$\overline{A} := \Omega \setminus A$$

complementary event

$$A \cap B = \emptyset$$

A and B are incompatible

Examples

One coin flip

$$\Omega = \{H, T\}$$

$$\mathcal{P}(\Omega) = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

Event "heads": $E = \{H\}$

Two coin flips

$$\Omega = \{HH, HT, TH, TT\}$$

Event "at least one head": $E = \{HH, HT, TH\}$

Four outcomes, sixteen possible events.

The Event Space Grows Fast

Three-sided die: $\Omega = \{1, 2, 3\}$, so $\mathcal{P}(\Omega)$ has $2^3 = 8$ events, ordered by inclusion:

$\{1, 2, 3\}$ $\{1, 2\}$ $\{1, 3\}$ $\{2, 3\}$ $\{1\}$ $\{2\}$ $\{3\}$ $\emptyset$

For $|\Omega| = n$ the event space has 2^n elements — which is exactly why the next unit is about counting.

Not Every Sample Space Is Finite

Dosage of a drug

$$\Omega = (0, \infty)$$

Uncountably many outcomes.

Race with 7 teams

$$\Omega = \{\text{all orderings of } 1, \dots, 7\}$$

$7! = 5040$ outcomes.

Choosing Ω is a modelling decision: it fixes what counts as an outcome, and therefore which questions can be asked at all.

Venn Diagrams

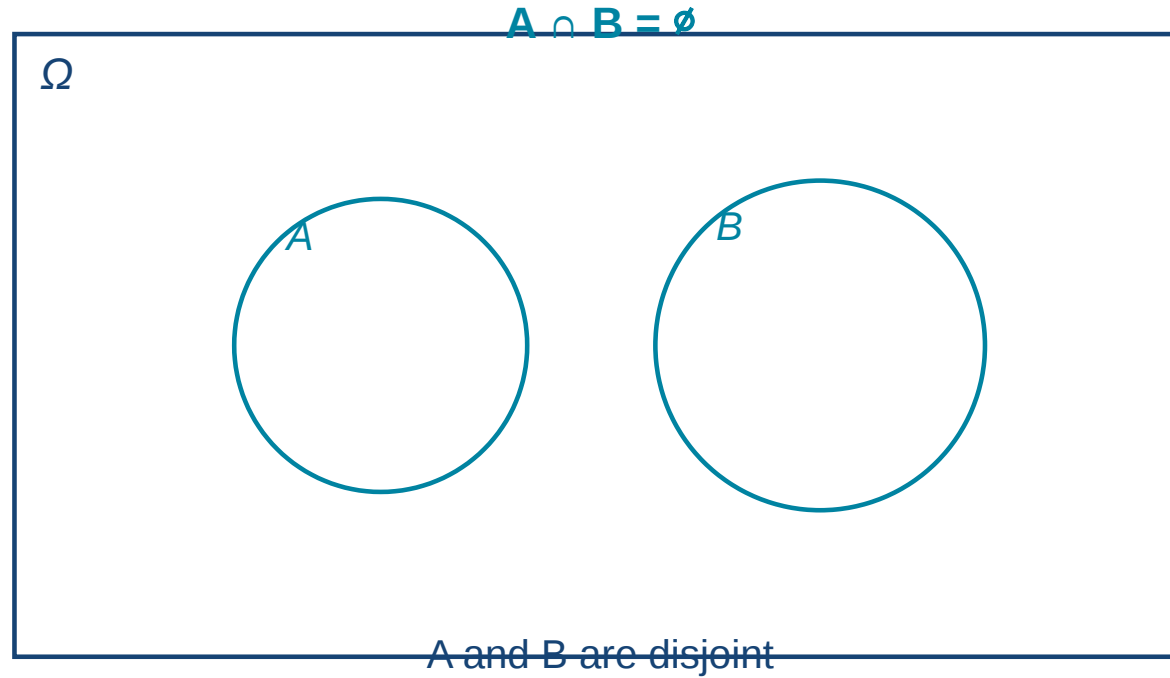
Venn diagrams are the graphical representation of events:

- the rectangle represents the sample space Ω
- circles represent events $A \subseteq \Omega, A \in \mathcal{P}(\Omega)$

Every region of the picture *is* a set — and every set operation is a region you can point at.

Important Event Types

Before each step: which region will light up?

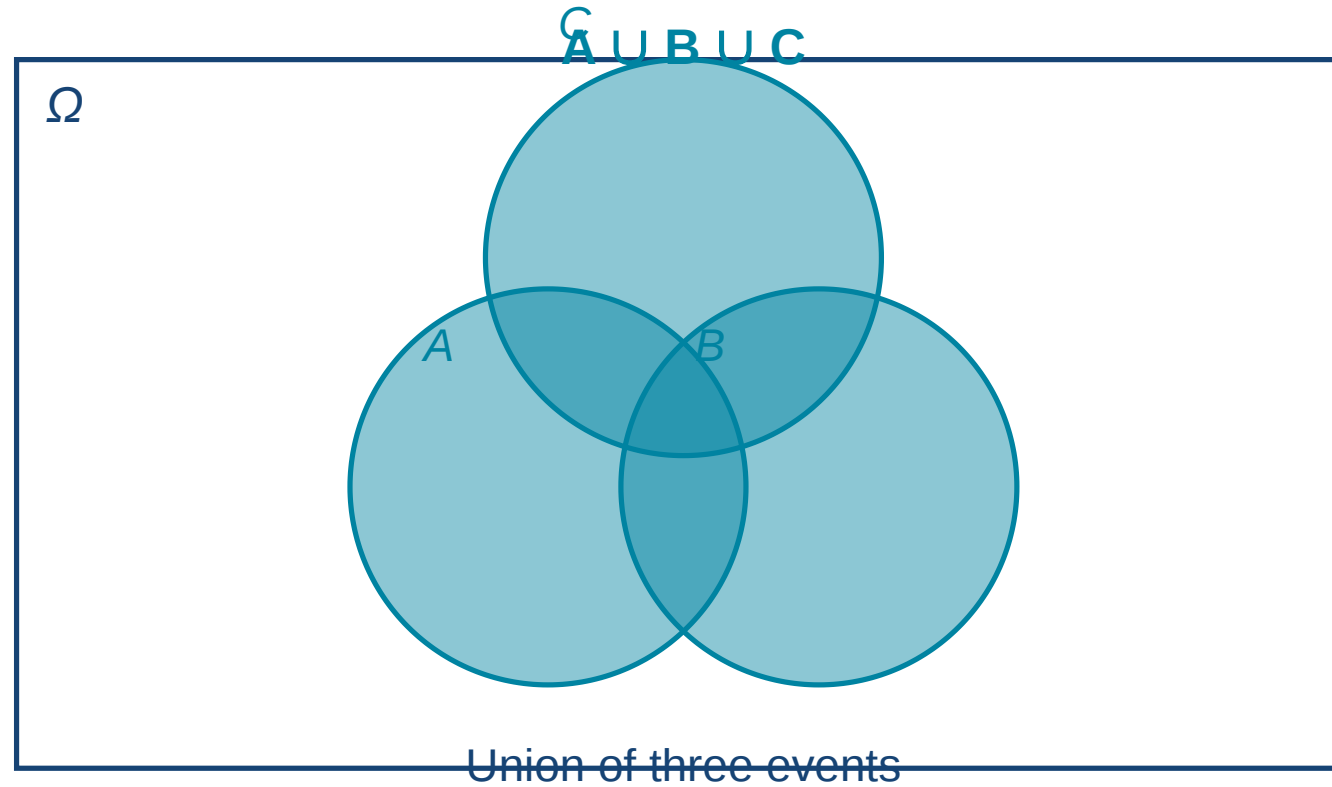


Algebra of Events

Law	Union form	Intersection form
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$	$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

Events form a set algebra — the same laws you know, but now every identity is a picture.

Set Operations as Regions



Mini-Exercise

Sketch (pen and paper) the regions for:

1. $(A \cup B) \cap C$
2. $(A \cap B) \cup C$

Are they the same region? Which algebra law says so — or says they are not?

Mini-Exercise — Solution

Mini-Exercise — Solution

They differ: $(A \cup B) \cap C$ lies entirely **inside** C , while $(A \cap B) \cup C$ contains **all** of C plus the lens $A \cap B$ outside it.

The distributive laws expand each into a different expression — the algebra distinguishes exactly what the picture shows.

An Intuition: Probability as Area

Draw a Venn diagram on the ground, with each area proportional to the number of elementary events it contains. Throw a stone into the rectangle at random.

The probability of A is how likely the stone lands inside A , assuming

1. we hit the rectangle Ω for certain, and
2. every part of Ω is equally likely to be hit.

Probability is then a **measure of the state** $A \subseteq \Omega$ of a random experiment — literally, how much of the sample space the event occupies.

Both assumptions are doing real work.
Assumption 2 is the one we will have to give up.

Two Ways to Read a Probability

Frequentist

The probability of an elementary event a is the fraction of repetitions of the experiment that yield a .

Requires the experiment to be repeatable.

Bayesian

The probability of a is the degree of certainty that a will happen — even without conducting the experiment.

Requires a stated state of knowledge.

Both interpretations yield the same mathematics. They are mental models for what the number *means*, not different theories.

The Classical (Laplace) Definition

Definition (Classical / Laplace probability). If every elementary event in Ω is assigned the same probability, then

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of cases favourable for } A}{\text{number of possible outcomes}}$$

An experiment whose configuration and procedure make outcomes equiprobable — by definition or by construction — is a **Laplace experiment** its probabilities need not be estimated, they are implied by the setup.

Why That Definition Is Not Enough

The circularity. "Laplace probability" defines probability using *equiprobability* — that is, using another notion of probability. Strictly speaking it is a circular definition.

The frequentist repair — defining probability as the limit of relative frequencies — fails too: such a limit exists only in mathematical settings, never in an experiment that can actually be repeated.

So what do we do? Stop trying to *define* what probability is, and instead specify what it must *do*.

The Kolmogorov Axioms

Definition (Probability measure, Kolmogorov 1933). Let Ω be a sample space and $\mathcal{P}(\Omega)$ its event space. A function $P : \mathcal{P}(\Omega) \rightarrow \mathbb{R}$ is a **probability measure** if:

Axiom I

$$P(A) \geq 0$$

non-negative

Axiom II

$$P(\Omega) = 1$$

normalised

Axiom III

$$A \cap B = \emptyset \Rightarrow$$

$$P(A \cup B) = P(A) + P(B)$$

additive

Nothing is said about *which* probabilities the events get — only about how any assignment must behave.

Beyond finite sample spaces. When Ω is infinite, the event space is not the full power set but a σ -**algebra** of measurable sets, and Axiom III is required for **countably many** disjoint

Probability Space

Definition. Let Ω be a sample space, $\mathcal{P}(\Omega)$ an event space, and P a probability measure. The triple $(\Omega, \mathcal{P}(\Omega), P)$ — also written (Ω, P) — is a **probability space**.

Every probabilistic statement in this course is made inside some probability space. Naming it is the discipline that keeps modelling assumptions visible.

What the Axioms Already Give Us

$$P(A) + P(\overline{A}) = 1$$

(Axioms II, III)

$$P(\emptyset) = 0$$

(from the above with $A = \Omega$)

$$A \subseteq B \Rightarrow P(A) \leq P(B)$$

monotonicity (Axioms I, II)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

inclusion–exclusion (Axiom III)

For mutually exclusive $A_1, \dots, A_p$: $P(A_1 \cup \dots \cup A_p) = P(A_1) + \dots + P(A_p)$.

Takeaway

Probability is not defined by counting or by repetition — it is any assignment of numbers to events that is **non-negative, normalised, and additive**. The Laplace formula is one such assignment, not the definition.

Module 2

Conditioning Is Inference

Guiding Question: *A 99%-accurate test says you are sick. Why is that not a 99% chance of being sick — and what exactly does the evidence do to your belief?*

Motivating Example

You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

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You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

- Before the first roll: $\Omega = \{(1, 1), (1, 2), \dots, (6, 6)\}$, $|\Omega| = 36$
- After seeing the 3, only $\Omega_{red} = \{(3, 1), \dots, (3, 6)\}$ remains
- Outcomes like $(2, 4)$ have become **impossible**
- The reduced space now carries all the probability: $P(\Omega_{red}) = 1$
- "Sum is 8" inside it: $E = \{(3, 5)\}$

$$P = \frac{1}{6}$$

Definition

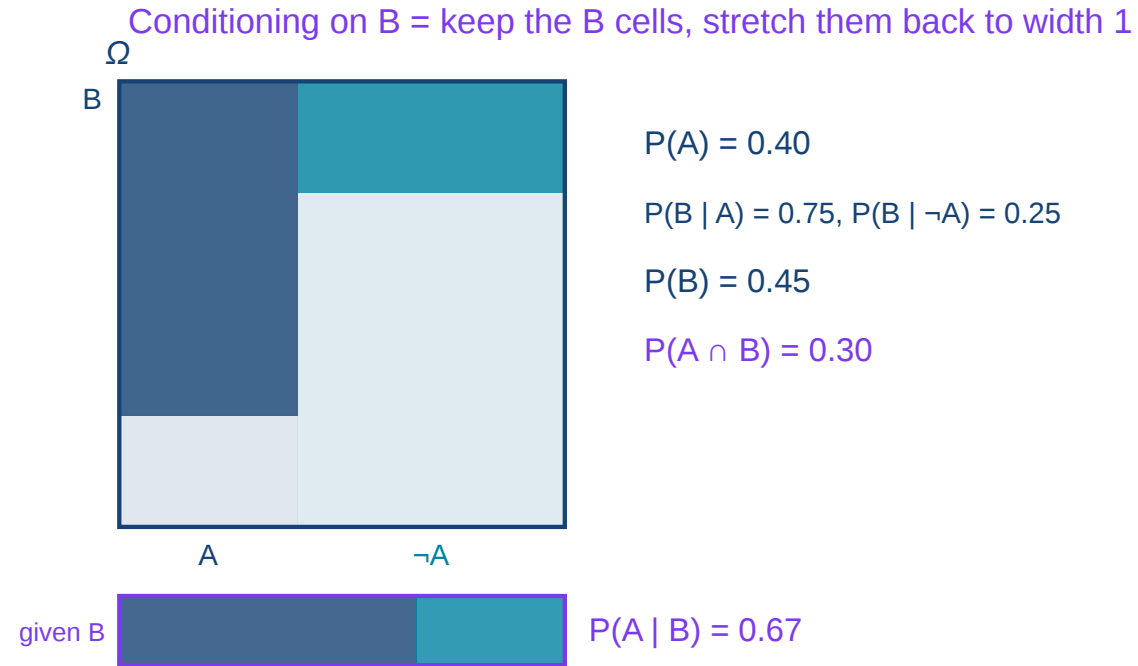
Definition (Conditional probability).

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad \text{if } P(B) > 0$$

$P(A \mid B)$ is the probability of A **under the condition** B .

Back to the stone: *given that the stone landed in area B , how likely is it that it also landed in A ?*
We renormalise by the area of B , because B is now the whole world.

Conditioning, Drawn



Before the last click: what must the two shaded cells be divided by?

Example: Two Coins

Two coins are flipped. The event of interest is $B = \{(h, h)\}$.

(a) "The first coin is heads"

$$F = \{(h, h), (h, t)\}$$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

(b) "At least one coin is heads"

$$A = \{(h, h), (h, t), (t, h)\}$$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Same event B , two different pieces of information, two different answers. **Why is (b) smaller?**

Total Probability

Theorem (Total probability). Let $B_1, \dots, B_k$ be mutually exclusive with $\Omega = B_1 \cup \dots \cup B_k$ and $P(B_i) > 0$. Then for any event A :

$$P(A) = \sum_{i=1}^k P(B_i) \cdot P(A \mid B_i)$$

Proof. Decompose A along the partition and use additivity:

$$P(A) = P((B_1 \cup \dots \cup B_k) \cap A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i) \cdot P(A \mid B_i)$$

Rules That Follow

Product rule

$$\begin{aligned} P(A \cap B) &= P(B) \cdot P(A \mid B) \\ &= P(A) \cdot P(B \mid A) \end{aligned}$$

Reversing the condition

$$P(B \mid A) = \frac{P(B) \cdot P(A \mid B)}{P(A)}$$

Complement, same condition

$$P(\overline{A} \mid B) = 1 - P(A \mid B)$$

But in general

$$P(A \mid \overline{B}) \neq 1 - P(A \mid B)$$

Held fixed at B , the map $A \mapsto P(A \mid B)$ satisfies the Kolmogorov axioms — conditioning yields a **new probability measure** P_B .

Chain Rule

$$P(E_1 \cap E_2 \cap \dots \cap E_n) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2) \cdots P(E_n \mid E_1 \cap \dots \cap E_{n-1})$$

By successive application of the product rule.

Any joint probability can be written as a chain of conditionals. Choosing a *good* order — and dropping conditions that do not matter — is exactly what graphical models will do later.

Takeaway

Conditioning **shrinks the sample space and renormalises**. Everything else — product rule, total probability, Bayes, the chain rule — is bookkeeping on that single move.

Independence

Definition (Statistical independence). Two events A and B are **independent** iff

$$P(A \cap B) = P(A) \cdot P(B)$$

If additionally $0 < P(B) < 1$, this is equivalent to

$$P(A \mid B) = P(A) \quad \text{and} \quad P(A \mid \overline{B}) = P(A)$$

Independence means: **learning B changes nothing about A .**

The Classic Confusion

Statistical independence does not mean the events are disjoint.

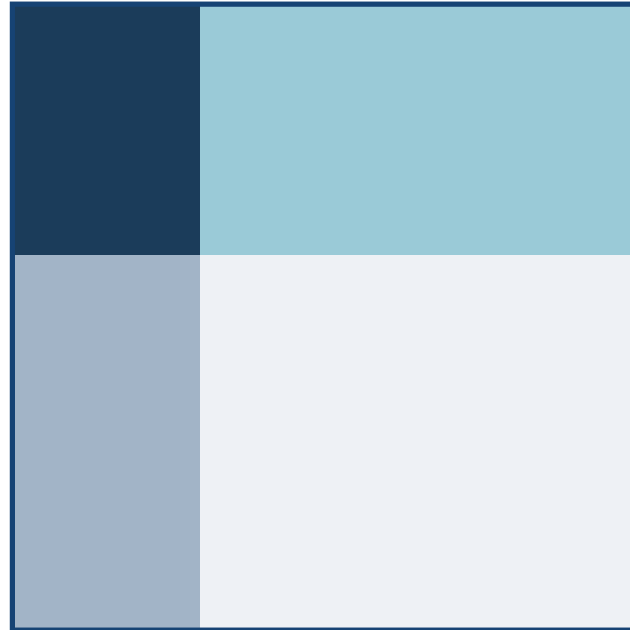
Disjoint events with positive probability are about as **dependent** as events can be: if A occurs, B certainly did not.

$$A \cap B = \emptyset \Rightarrow P(A \mid B) = 0 \neq P(A)$$

Independence Is About Areas, Not Overlap

$$0.3 \cdot 0.4 = 0.12 = P(A \cap B) \rightarrow \text{independent}$$

Ω

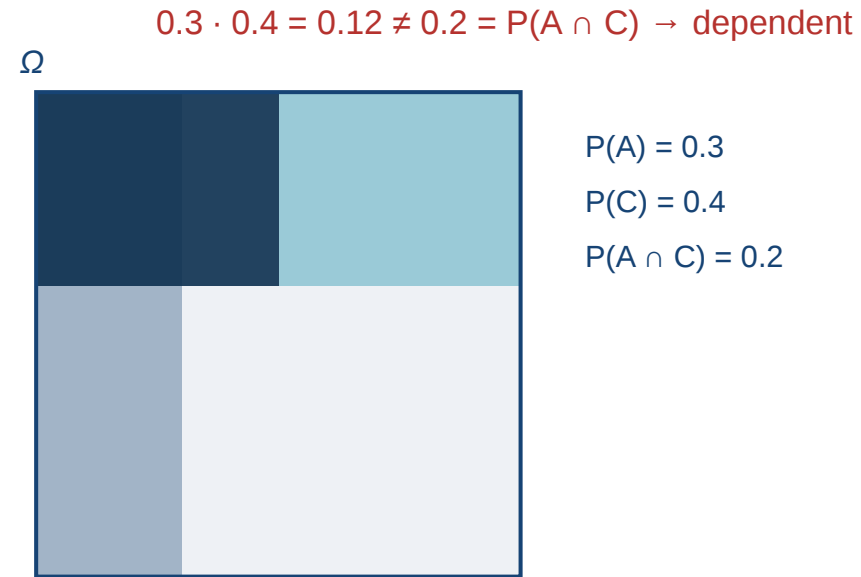


$$P(A) = 0.3$$

$$P(B) = 0.4$$

$$P(A \cap B) = 0.12$$

The Same Picture, Dependent



Here $P(A) \cdot P(C) = 0.12$ but $P(A \cap C) = 0.20$: the intersection is **larger** than independence would predict — A and C occur together more often than chance.

More Than Two Events

Events $A_1, \dots, A_n$ are (mutually) independent iff for **every** subset of indices $i_1, \dots, i_l$:

$$P(A_{i_1} \cap \dots \cap A_{i_l}) = P(A_{i_1}) \cdots P(A_{i_l})$$

Pairwise independence is **not** enough — every subset must factorise.

Takeaway

Independence is a statement about **numbers**, not pictures: $P(A \cap B)$ equals the product of the marginals. Disjoint is not independent, and identical marginals can hide either.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

Bayes' Theorem — An Intuition First

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Your test comes back positive. How likely is it that you are actually sick?

Only about **9%** — the answer most people miss by a factor of ten.

Why? Count People, Not Percentages

Restricting to the positives is exactly the conditioning step

$$P(\text{sick} \mid \text{positive}) = 1/11 \approx 9\%$$



What Just Happened

Restricting attention to the people who tested positive is exactly the move from the conditional-probability definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The 11 positives are the new sample space B the single sick person among them is $A \cap B$. The rarity of the disease survives the evidence because there were so many more healthy people to misclassify.

The trap

"99% accurate" describes $P(\text{positive} \mid \text{sick})$.

You wanted $P(\text{sick} \mid \text{positive})$.

These are different numbers.

Bayes' Theorem — Two Events

Theorem (Bayes, 2 events). Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

posterior — what we believe after
likelihood evidence well explains B
prior — belief before the evidence

Which of the four was "99% accurate", and which one did the intuition ignore?

Proof — Two Events

From the definition of conditional probability, twice:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{P(A \cap B)}{P(A)} \cdot P(A)}{P(B)} = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Bayes' Theorem — Multiple Hypotheses

Theorem (Bayes, multiple events). Let $A_i \in \mathcal{P}(\Omega)$ be mutually exclusive with $\Omega = A_1 \cup \dots \cup A_k$ and $P(A_i) > 0$, and let B be an event with $P(B) > 0$. Then

$$P(A_i | B) = \frac{P(B | A_i) \cdot P(A_i)}{P(B)} = \frac{P(B | A_i) \cdot P(A_i)}{\sum_{j=1}^k P(B | A_j) \cdot P(A_j)}$$

The denominator is the **total probability** of the evidence: the same observation, explained by every competing hypothesis, weighted by how much we believed each one beforehand.

Proof — Multiple Hypotheses

As before, with the theorem of total probability supplying the denominator:

$$P(B) = \sum_{j=1}^k P(B \mid A_j) \cdot P(A_j)$$

Mini-Exercise

Same test, but now screening a **high-risk group** where 1 in 20 people is sick — the test is unchanged.

Estimate $P(\text{sick} \mid \text{positive})$ before computing it.

Mini-Exercise — Solution

Mini-Exercise — Solution

Out of 1,000: 50 sick $\rightarrow \approx 50$ true positives 950 healthy $\rightarrow \approx 10$ false alarms.

$$P(\text{sick} \mid \text{positive}) \approx \frac{50}{60} \approx 83$$

The test did not change. The prior did. This is why screening decisions depend on who is screened.

Takeaway

Bayes' theorem is the arithmetic of changing your mind: **posterior** $\propto$ **likelihood** $\times$ **prior**.
Evidence never speaks alone — it is always weighed against how plausible the hypothesis was to begin with.

Module 3

From Outcomes to the Joint

Guiding Question: *We rarely care which outcome occurred, only what it was worth — and never about one quantity alone. How do we get from a sample space to the joint distribution the three moves start from?*

Motivation: We Rarely Care About the Outcome Itself

Toss three fair coins: $\Omega = \{H, T\} \times \{H, T\} \times \{H, T\}$, eight outcomes.

Usually the interesting quantity is not *which* sequence occurred, but the **number of heads** Y :

$P(Y = 0) = \frac{1}{8}$	$P(Y = 1) = \frac{3}{8}$	$P(Y = 2) = \frac{3}{8}$	$P(Y = 3) = \frac{1}{8}$
(T, T, T)	$(H, T, T), (T, H, T), (T, T, H)$	$(H, H, T), (T, H, H), (H, T, H)$	(H, H, H)

Y compresses eight outcomes into four numbers — and the probabilities come along automatically.

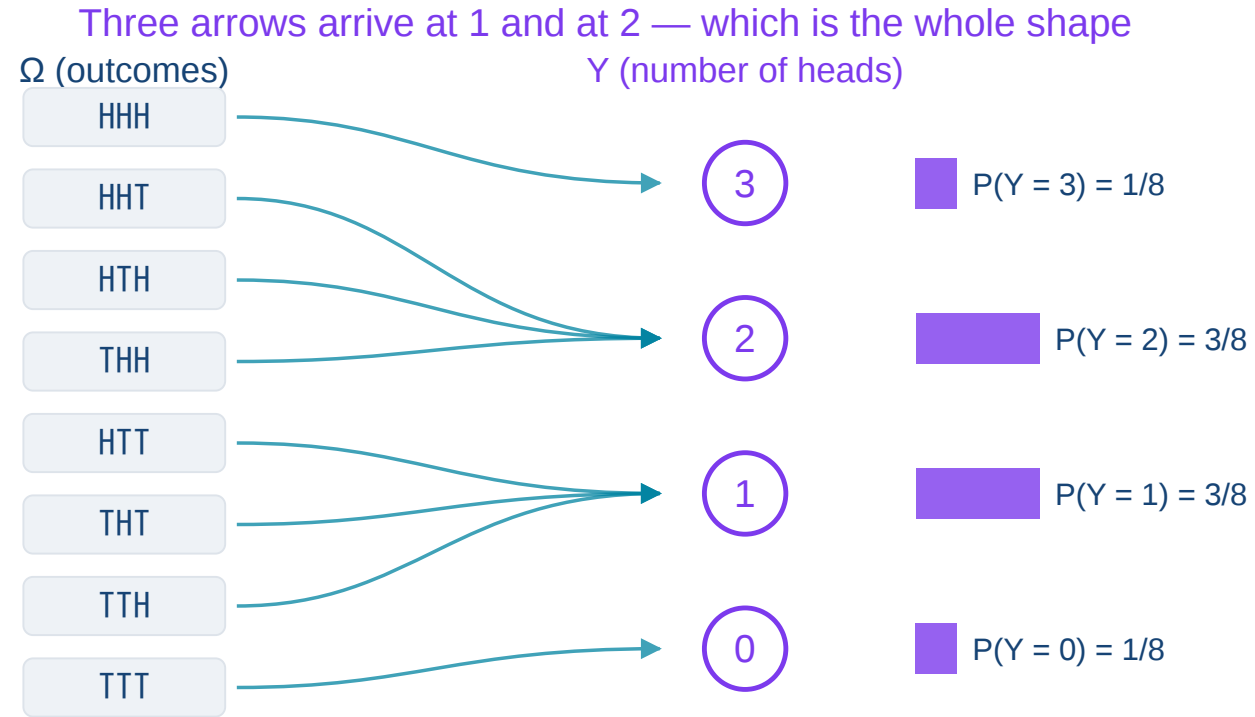
Definition

Definition (Random variable). A random variable X maps outcomes to a measurable range E : $X : \Omega \rightarrow E$. Its probabilities are inherited from the underlying probability space:

$$P(X \in S) = P(\{\omega \in \Omega \mid X(\omega) \in S\})$$

A random variable is neither random nor a variable — it is a **function** on the sample space. The randomness lives in ω , not in X .

The Function, Drawn



Examples

Two coins

$$\Omega = \{HH, HT, TH, TT\}$$

X = number of heads

People

$$\Omega = \{\text{person}_1, \text{person}_2, \dots\}$$

X = height of the person

Two dice

$$\Omega = \{\langle n_1, n_2 \rangle\}$$

$$X(\omega) = n_1 + n_2$$

Discrete and Continuous

Discrete random variables

Countable range, usually $E \subseteq \mathbb{N}$.

- number of children of a person
- sum of two dice

Continuous random variables

Real-valued range, $E = \mathbb{R}$.

- height of a person
- horsepower of a car

The distinction decides *how* we may describe the probabilities — the subject of the next two units.

A Random Variable With Infinitely Many Values

Flip a biased coin ($P(H) = p$) until the first head. Let X be the number of flips needed:

$$P(X = 1) = p, \quad P(X = 2) = (1 - p)p, \quad P(X = 3) = (1 - p)^2 p$$

$$P(X = n) = (1 - p)^{n-1} p$$

Countably infinite range, and the probabilities still sum to 1.

Takeaway

A random variable is a **function from outcomes to numbers** that carries the probability measure with it. It lets us stop talking about sample spaces and start talking about quantities — which is what the rest of the course does.

Probability Mass Function

Definition (PMF). For a discrete random variable X , the **probability mass function** is

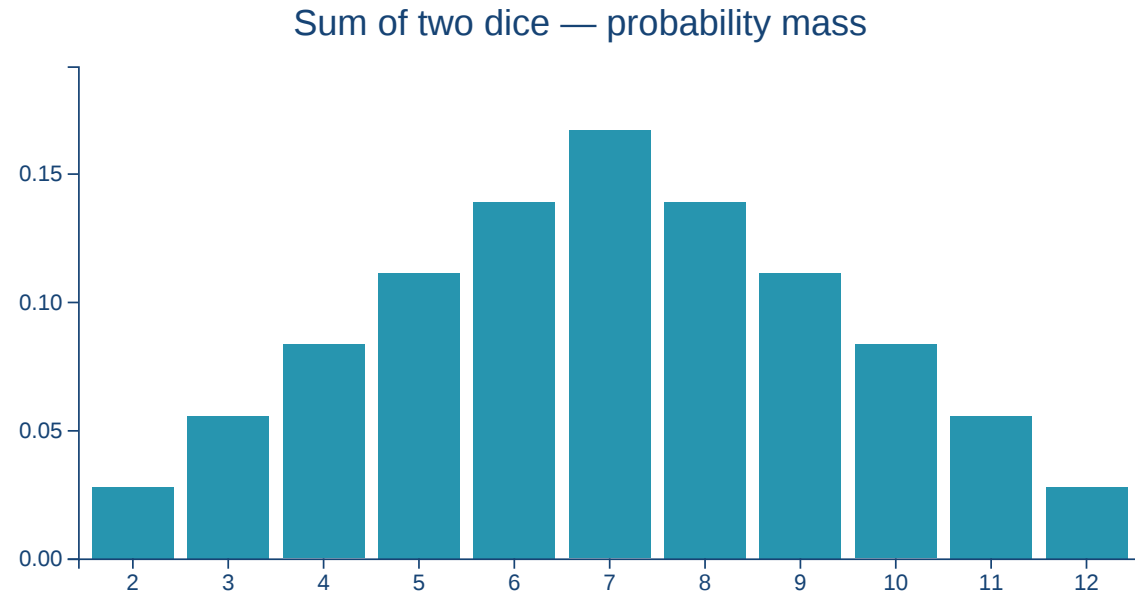
$$p(a) = P(X = a)$$

$p(a) > 0$ for at most countably
many a

$$\sum_i p(a_i) = 1$$

p is itself a probability measure

Example: Sum of Two Dice



The triangular shape is the counting from earlier units, drawn: six of the 36 outcomes give 7, only one gives 2.

Expectation

Definition (Expected value).

$$E[X] = \sum_i a_i \cdot p(a_i)$$

- Frequentist reading: the **average outcome** if the experiment were repeated infinitely often
- The first moment: $\mu = E[X^1]$ the first *central* moment $E[X - E[X]]$ is always 0

For a real function g :

$$E[g(X)] = \sum_i g(a_i)p(a_i)$$

and for constants a, b :

$$E[aX + b] = aE[X] + b$$

Variance

Definition (Variance).

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$$

- a measure of **dispersion** around the mean — larger variance, larger typical deviation
- the second central moment
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- $\sigma = \sqrt{\text{Var}(X)}$ is the **standard deviation**, in the same unit as X

Bernoulli and Binomial

Bernoulli(p) — one trial

$X = 1$ success, $X = 0$ failure

$$E[X] = p, \quad \text{Var}(X) = p(1 - p)$$

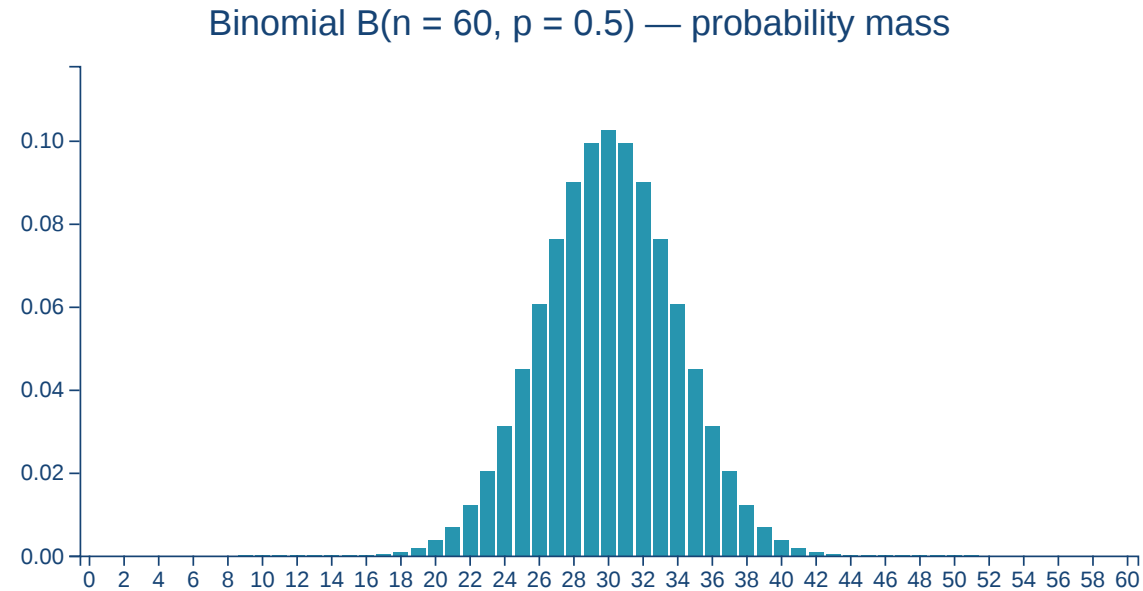
Binomial $B(n, p)$ — n independent trials

$$p(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$E[X] = np, \quad \text{Var}(X) = np(1 - p)$$

The binomial PMF is the counting formula times the probability of one specific sequence — combinatorics and independence in a single line.

How the Shape Depends on the Parameters



Predict before each click: as n grows, does the distribution get wider or narrower — and what happens to its shape?

Mini-Exercise

A quiz has 10 questions with 4 options each. A student guesses every answer.

1. Which distribution describes the number of correct answers?
2. What is the expected number correct?
3. Is passing (≥ 5 correct) likely?

Mini-Exercise — Solution

Mini-Exercise — Solution

1. $B(10, 0.25)$ — ten independent trials, constant success probability
2. $E[X] = 10 \cdot 0.25 = 2.5$
3. $P(X \geq 5) \approx 0.078$ — under 8%

The independence assumption is what makes this a binomial ask when it would fail in a real exam.

Takeaway

A discrete distribution is a PMF expectation and variance compress it into **where it sits** and **how much it spreads**. Bernoulli and binomial are the counting units in probabilistic clothing.

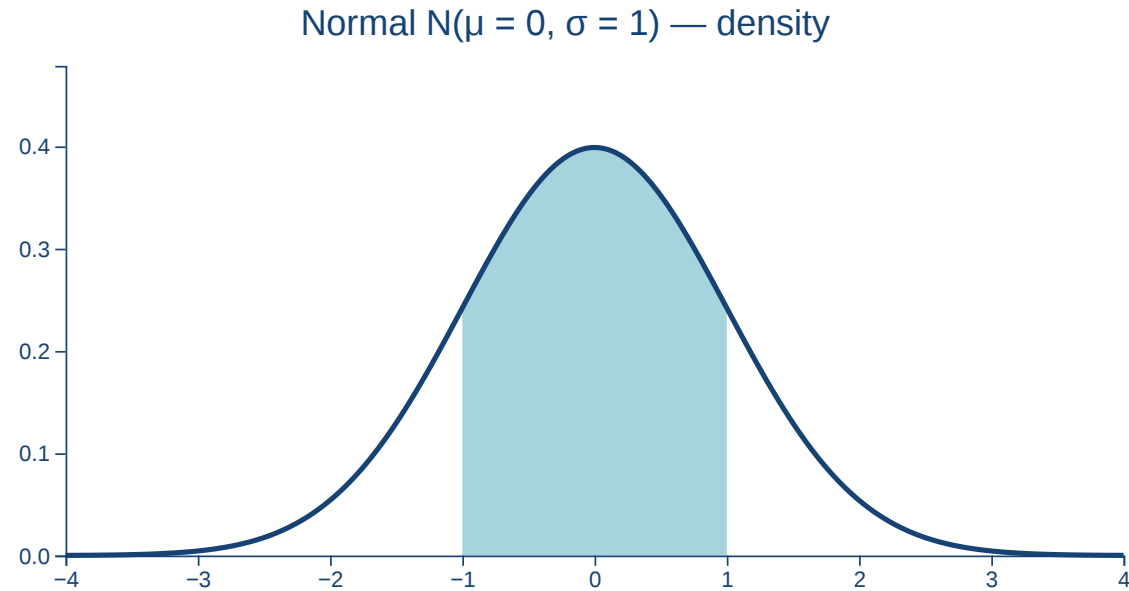
From Mass to Density

Definition (PDF). A continuous random variable X has a **probability density function** f with

$$P(b \leq X \leq c) = \int_b^c f(a)da, \quad \int_{-\infty}^{\infty} f(a)da = 1$$

$f(a)$ is **not** a probability — it is probability *per unit interval*. Consequently $f(a) > 1$ is perfectly possible, and $P(X = a) = 0$ for every single value a .

Probability Is Area Under the Curve



The shaded region is $P(-1 \leq X \leq 1)$ — the same "probability as area" intuition from the axioms unit, now with a curve instead of a Venn diagram.

Expectation and Variance — Sums Become Integrals

Discrete

$$E[X] = \sum_i a_i p(a_i)$$

$$E[g(X)] = \sum_i g(a_i) p(a_i)$$

Continuous

$$E[X] = \int_{-\infty}^{\infty} a f(a) da$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(a) f(a) da$$

The rules are unchanged: $E[aX + b] = aE[X] + b$, and $\text{Var}(X) = E[(X - \mu)^2]$.

Two Basic Continuous Distributions

Uniform on $[a, b]$

$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$

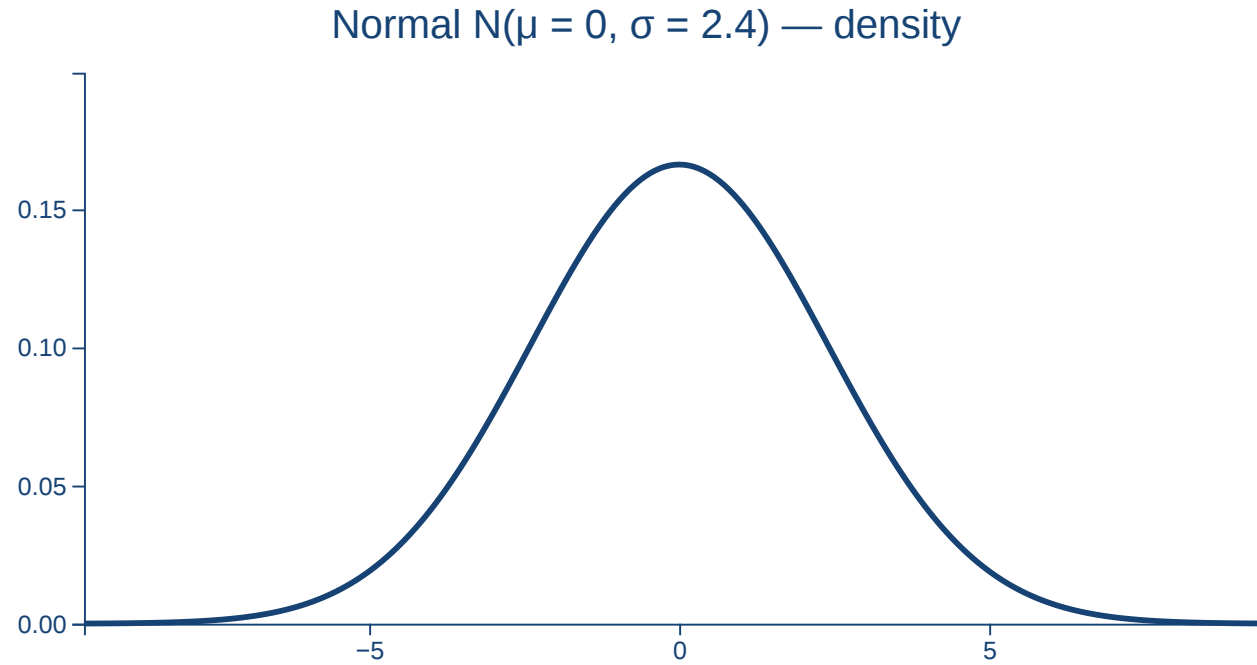
Every interval of equal length is equally likely — the Laplace assumption, continuous version.

Normal $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$E[X] = \mu$, $\text{Var}(X) = \sigma^2$; $N(0, 1)$ is the **standard normal**.

What σ Does to the Shape



The peak falls as the curve widens — the area is always 1.

Why the Normal Is Everywhere

The **central limit theorem**: under mild conditions, averages of many independent observations converge to a normal distribution — whatever the original distribution was.

This is why the normal is the default when the true distribution is unknown, and why the binomial started looking bell-shaped as n grew in the previous unit.

Other Distributions You Will Meet

- **Exponential** — time until an event occurs (e.g. until the next earthquake)
- **Gamma** — time until n events have occurred
- **Beta** — values confined to a finite interval $[a, b]$
- **Dirichlet** — distributions over probabilities

Continuous distributions replace mass with **density** and sums with **integrals**. Probability remains area, expectation remains a weighted average, and the normal is the default shape because of the central limit theorem.

More Than One Random Variable

Univariate

one random variable

Bivariate

two random variables

Multivariate

three or more

A **joint distribution** describes the outcomes of several random variables *at the same time* — which is what every dataset with more than one column is.

Joint Mass and Density

Discrete — joint PMF

$$p_{X,Y}(a,b) = P(X = a, Y = b)$$

Continuous — joint PDF

$$P((X,Y) \in R) = \iint_R f_{X,Y}(a,b) da db$$

Notational convention from here on. We stop distinguishing discrete from continuous variables and simply write $p(a,b,c,\dots)$ for the probability that $\{X_1, X_2, X_3, \dots\}$ is in state $a, b, c, \dots$

Marginalisation

Marginalisation eliminates a random variable by summing over all its states. For $p(x_1, \dots, x_n)$, a **marginal** over a subset is

$$p(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, x_2, \dots, x_n)$$

"Forget about a variable, but keep its influence correctly accounted for."

Example: Boys and Girls

15% of families have no children, 20% have one, 35% have two, 30% have three. Each child is independently a boy or a girl with probability $\frac{1}{2}$.

Let B = number of boys, G = number of girls.

$p(B, G)$	$G = 0$	$G = 1$	$G = 2$	$G = 3$	$p(B)$
$B = 0$	.15	.10	.0875	.0375	.375
$B = 1$	.10	.175	.1125	0	.3875
$B = 2$	.0875	.1125	0	0	.2
$B = 3$	.0375	0	0	0	.0375
$p(G)$	.375	.3875	.2	.0375	1

The margins of the table *are* the marginals — which is where the name comes from.

Covariance

$$\text{cov}(A, B) = E[(A - E[A])(B - E[B])]$$

$$\text{cov} > 0$$

A large when B large —
positively related

$$\text{cov} < 0$$

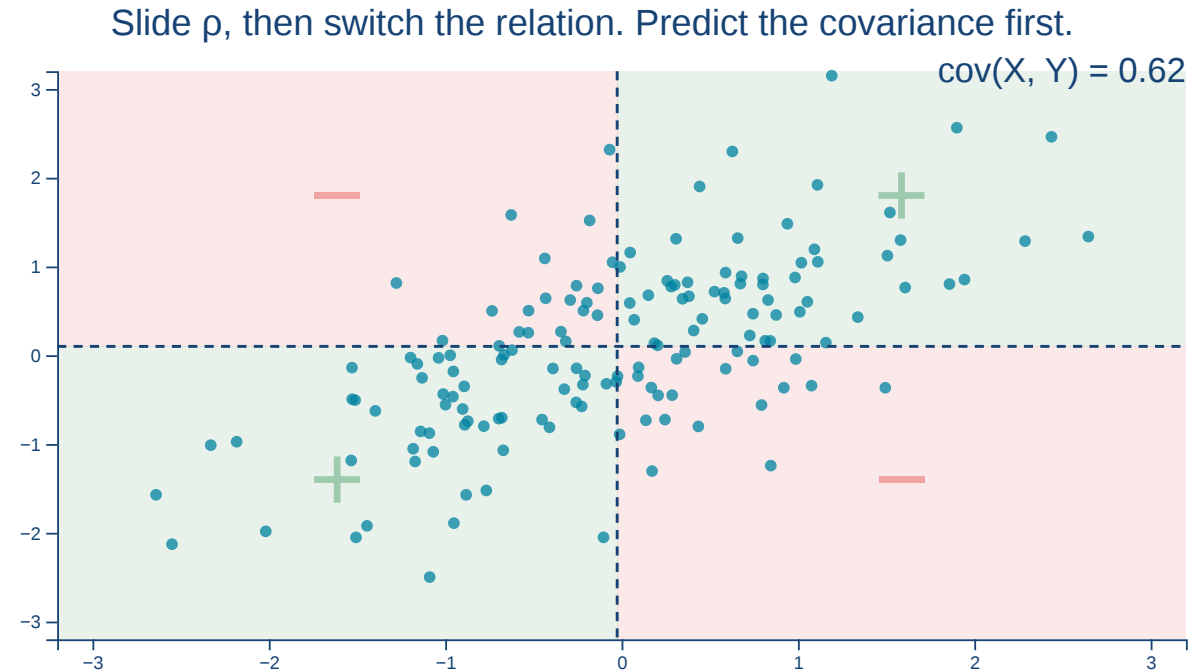
A large when B small —
negatively related

$$\text{cov}(A, A) = \text{Var}(A)$$

variance is self-covariance

Independence implies zero covariance. **The converse is false** — covariance only detects *monotone linear* association.

Covariance Sees Only One Kind of Structure



Predict first: with $Y = X^2 - 1$, every point is fixed by its x . The covariance?

Takeaway

The joint distribution holds everything **marginalisation** sums a variable away, **conditioning** restricts, and **covariance** summarises linear co-variation. Zero covariance is weaker than independence.

Module 4

The Data Is a Sample

Guiding Question: *The distribution is unknown and all we ever have is a sample. What can we honestly say about the population — and how wrong are we allowed to assume we are?*

We Never Know the Distribution

Up to now we assumed a distribution was given. In practice it is **unknown**, and we must estimate its properties from data.

Population

The entire group — all humans on earth, all transactions, all documents.

Its properties (μ, σ^2) are the truth we cannot observe.

Sample

The observations we actually have.

Its properties $(\bar{x}, s^2)$ are what we compute — estimates of the population's.

Descriptive statistics summarise a sample, usually while assuming something about the underlying distribution.

What We Summarise

Central tendency

1st order moment
sample mean, median

Dispersion

2nd central moment
variance, standard deviation, IQR

Shape

3rd central moment
skewness, quantiles

Univariate statistics describe one random variable **multivariate** statistics describe $X_1, \dots, X_d$ together.

Means

Sample mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Weighted mean

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$$

Geometric mean — for rates and ratios

$$\left(\prod_i x_i \right)^{1/n}$$

Harmonic mean — for averaging rates

$$\frac{n}{\sum_i 1/x_i}$$

Trimmed mean: drop the k smallest and largest values, then average — a cheap way to buy robustness.

Median

The **median** is the middle value of the sorted sample: for odd n the $\frac{n+1}{2}$ -th value, for even n the average of the two middle values.

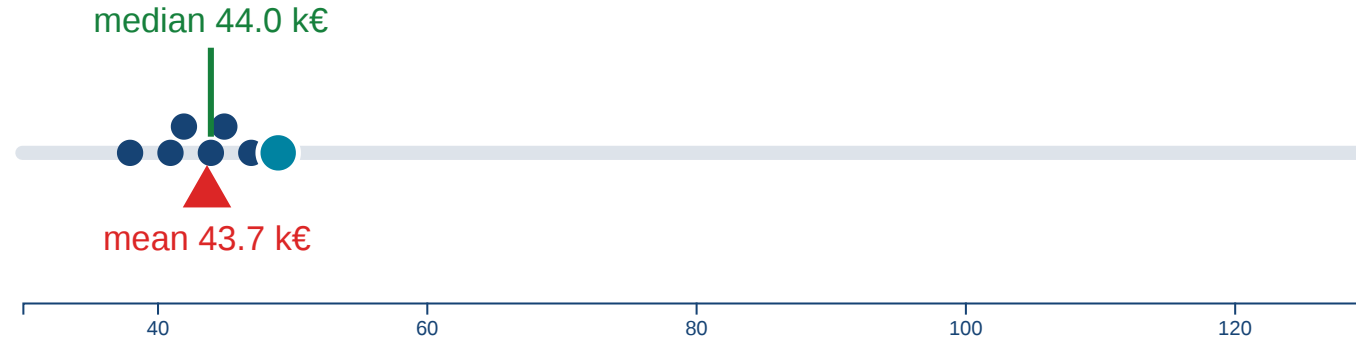
Robust against outliers — one extreme value moves the median by at most one position

Costlier to compute: requires sorting, or an interval approximation over binned data

For binned data, the median is interpolated within its interval from the cumulative counts, the interval's lower border and width.

One Value, Two Very Different Reactions

Drag the light point — which marker follows it?



Commit first: drag the top salary to the far right. Which marker follows?

Mini-Exercise

Salaries in a small firm (in k€): 38, 41, 42, 44, 45, 47, 260.

Compute the mean and the median. Which number should appear in a recruitment ad, and which one describes a typical employee?

Mini-Exercise — Solution

Mini-Exercise — Solution

Mean ≈ 73.9 , median = 44.

The single large value pulls the mean above **every** salary except one. The median describes the typical employee the mean describes the payroll divided by headcount. Both are correct — they answer different questions.

Dispersion

Dispersion describes how stretched or squeezed the data is around its centre.

Sample variance — the variance *of the observed data*

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Standard deviation $s = \sqrt{s^2}$, in the unit of the data.

Computable in **one pass** over the data, $O(n)$, using

$$\sum x_i^2 \text{ and } \sum x_i$$

— which matters when the data does not fit in memory.

Why the Denominator Changes

The **population variance** σ^2 is the variance over the whole population. If the true mean μ is unknown, we must estimate it — and using $\bar{x}$ in its place makes the deviations systematically too small.

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Dividing by $n - 1$ compensates: one degree of freedom was spent estimating the mean from the same data.

Takeaway

s^2 describes **the sample** $\hat{\sigma}^2$ estimates **the population**. The $n - 1$ is not a convention — it is the price of having estimated the mean from the same data.

Quantiles

The k -th q -**quantile** is the value x such that k/q of the values are smaller and $(q - k)/q$ are larger ($0 < k < q$).

Percentiles

$$q = 100$$

"the 12th percentile"

Quartiles

$$q = 4$$

$Q_1 = 25\text{th}$, $Q_2 = 50\text{th}$ (= median),
 $Q_3 = 75\text{th}$ percentile

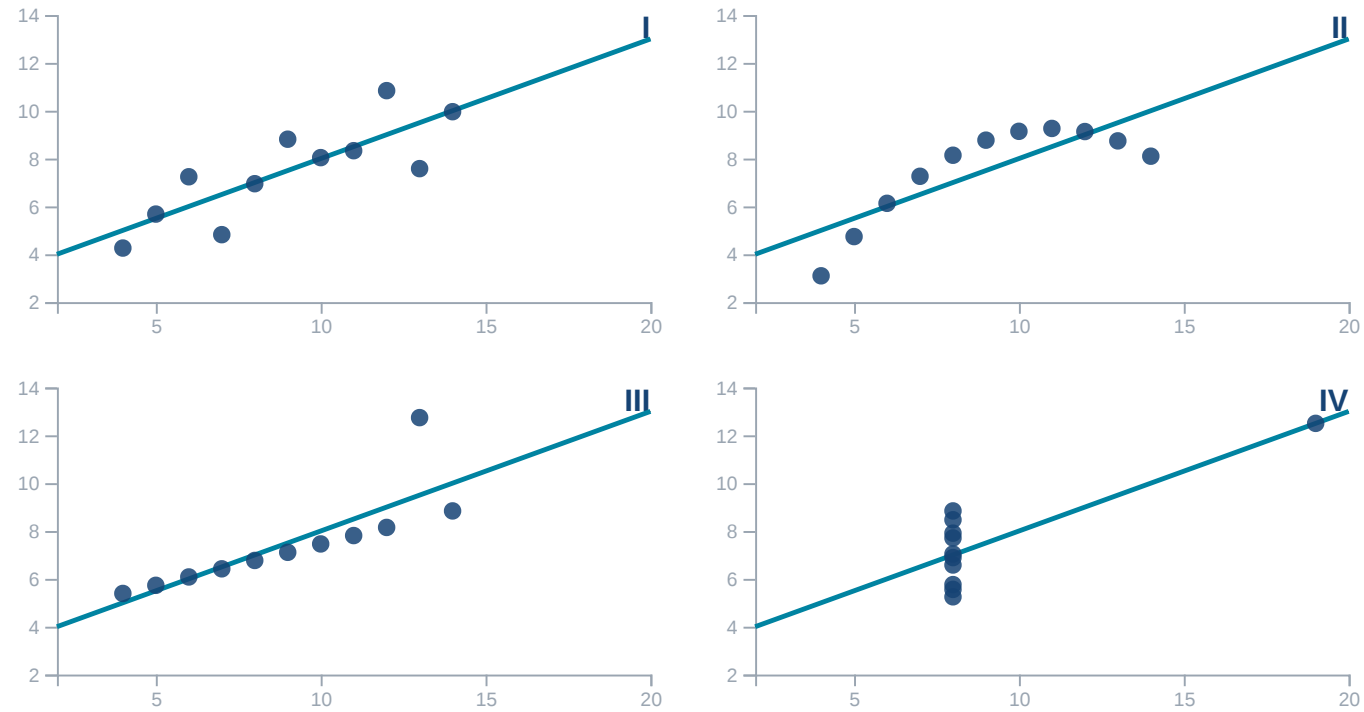
IQR

$$\text{IQR} = Q_3 - Q_1$$

the middle half of the data

Quantiles summarise **shape and skewness** without assuming a distribution.

Anscombe's Quartet



Same mean, same variance, same correlation, same regression line — different data

What the Quartet Teaches

Summaries compress — and compression loses.

Identical moments can hide a curve, an outlier, or a variable that barely varies at all.

Plot first, summarise second.

Visual inspection is not a soft alternative to statistics it is how you find out which statistics are meaningful.

Quantiles describe a distribution without assuming its shape — and no set of summary numbers can replace looking at the data.

You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of $\bar{x} = 43.7$ k€.

What is the probability that the population mean μ is exactly 43.7 k€?

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Zero. $\bar{x}$ is not μ — it is one **estimate** of it, computed from the seven people who happened to be in the sample.

The honest question is not *what is μ* but **how far from μ can this number be?**

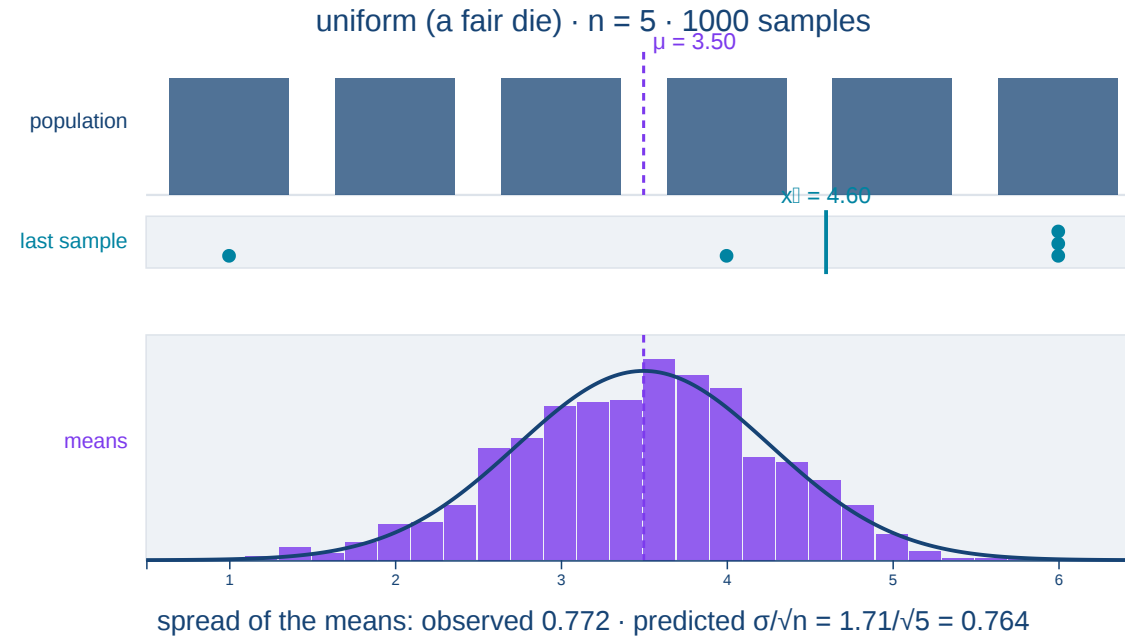
Estimator and Estimate

Definition. An **estimator** is a function of the sample — $\hat{\theta} = g(X_1, \dots, X_n)$ — used to approximate an unknown quantity θ of the population. The number it takes on for one particular sample is an **estimate**.

The sample is random, so the estimator is a **function of random variables**: $\hat{\theta}$ is itself a random variable, with a distribution of its own.

$\bar{x}$, s^2 and $\hat{\sigma}^2$ from the previous units are all estimators. We used them without saying what kind of object they are.

An Estimator Has a Distribution



Predict first: after a thousand samples, what shape will the lower picture have — and how wide?

Three Things the Picture Shows

- The means pile up **around** μ — the estimator is not systematically off
- The pile is **narrower than the population** — averaging cancels noise
- The pile is **bell-shaped**, even though the population was not

Definition (Sampling distribution). The distribution of an estimator $\hat{\theta}$ over all possible samples of size n . $\hat{\theta}$ is **unbiased** if $E[\hat{\theta}] = \theta$.

Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

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Which of the three observations survive a population that is not symmetric?

All three. The shape of the sampling distribution stops depending on the shape of the population as soon as n is more than a handful — which is why one formula can cover cases we never inspect.

Standard Error

Definition (Standard error). The standard deviation of an estimator's sampling distribution. For the sample mean of independent draws:

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

It is measured in the unit of the data and answers "how far off is a typical estimate", where σ answers "how far apart are typical *observations*".

σ and SE are different quantities. More data does nothing to σ — the world stays as variable as it was — and shrinks SE.

Precision Costs Data

Multiply the data by...	...and the standard error is multiplied by
2	0.71
4	$\frac{1}{2}$
25	$\frac{1}{5}$
100	$\frac{1}{10}$

The $\sqrt{n}$ is why the last digit of an estimate is so expensive: halving the error costs four times the data, and a tenfold improvement costs a hundredfold.

Two Theorems We Have Already Seen Happen

Law of large numbers

$\bar{X}_n \rightarrow \mu$ as $n \rightarrow \infty$.

The estimate converges on the truth —
the pile keeps narrowing.

Central limit theorem

$\bar{X}_n$ is approximately $N(\mu, \sigma^2/n)$ for large n .

And we know the *shape* of what is left.

The first says the error goes away; the second says how much of it is left after n observations. Only the second lets us put a number on our ignorance.

From a Number to an Interval

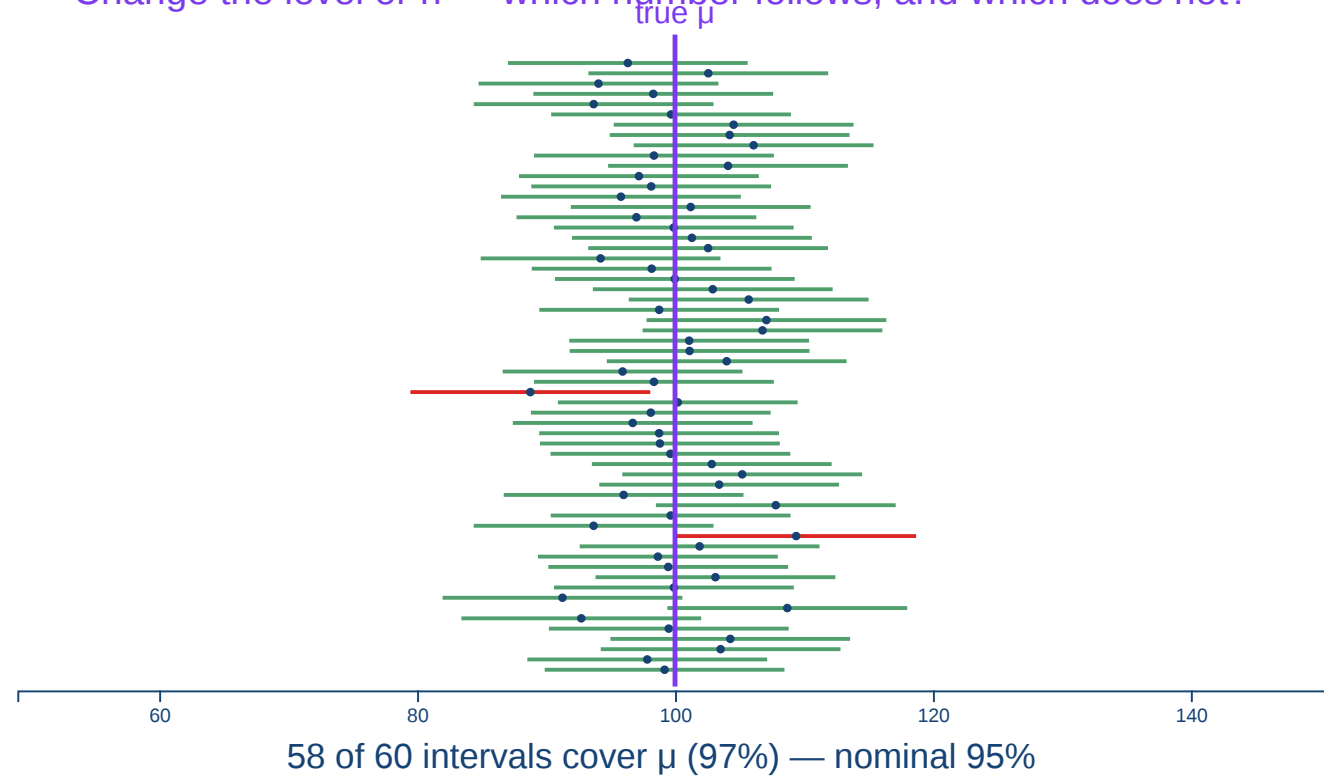
If $\bar{X} \approx N(\mu, \sigma^2/n)$, then for 95% of samples $\bar{X}$ lands within 1.96 SE of μ — and therefore μ lands within 1.96 SE of $\bar{X}$.

$$\bar{x} \pm 1.96 \cdot \frac{\sigma}{n}$$

the estimate — our best single guess
how confident we insist on being
how messy the estimate is

What Does "95% Confident" Mean?

Change the level or n — which number follows, and which does not?



The Sentence Almost Everyone Gets Wrong

WHAT IT DOES NOT MEAN

"There is a 95% probability that μ lies in **this** interval."

μ is a fixed number. It is either in this interval or it is not — there is no probability left to have.

WHAT IT MEANS

"95% of the intervals produced **by this procedure** contain μ ."

The probability is a property of the recipe, evaluated before the data is seen.

Where This Leaves Us

The introduction distinguished two kinds of uncertainty. We can now measure one of them.

- σ — how variable the world is. **Aleatoric**. Unchanged by data.
- $SE = \sigma / \sqrt{n}$ — how badly we know our own estimate.
Epistemic. Shrinks with data.

"More data helps" was a slogan.

It is now a rate: $1/\sqrt{n}$.

Mini-Exercise

A shop measures the time customers spend at a counter. From $n = 100$ customers: $\bar{x} = 180$ s, and past records give $\sigma = 60$ s.

1. What is the standard error of the mean?
2. Give an approximate 95% confidence interval.
3. The manager wants the interval to be **half** as wide. How many customers?

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1. $SE = 60/\sqrt{100} = 6$ s

2. $180 \pm 1.96 \cdot 6 \approx [168, 192]$ s

3. Half the width needs twice the $\sqrt{n}$, so **four times the data**: $n = 400$.

Note what did *not* change in part 3: $\sigma = 60$ s. Customers stay as variable as they were — only our knowledge of their average improves.

Takeaway

A statistic is a **random variable**. Its distribution over repeated samples is the sampling distribution; its spread is the **standard error**, $\sigma/\sqrt{n}$. A confidence interval reports that spread — and its percentage describes the **procedure**, not the one interval in front of you.